

# Object Oriented Design Quality Metrics

## References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [jqassistant](#)
- [notebook walks through examples for integrating various packages with Neo4j](#)
- [OO Design Quality Metrics](#)
- [A Validation of Martin's Metric](#)
- [Neo4j Python Driver](#)

## Incoming Dependencies

Incoming dependencies are also denoted as "Fan-in", "Afferent Coupling" or "in-degree". These are the ones that use the listed package.

If these packages get changed, the incoming dependencies might be affected by the change. The more incoming dependencies, the harder it gets to change the code without the need to adapt the dependent code ("rigid code"). Even worse, it might affect the behavior of the dependent code in an unwanted way ("fragile code").

Since Java Packages are organized hierarchically, incoming dependencies can be count for every package in isolation or by including all of its sub-packages. The latter one is done without top level packages like for example "org" or "org.company" by assuring that only packages are considered that have other packages or types in the same hierarchy level ("siblings").

### Table 1a

- Show the top 20 Java Packages with the most incoming dependencies
- Set the "incomingDependencies" properties on Package nodes.

|    | artifactName             | fullQualifiedPackageName                          | packageName     | incomingDependencies | incomingDependenciesWeight | incon |
|----|--------------------------|---------------------------------------------------|-----------------|----------------------|----------------------------|-------|
| 0  | axon-messaging-5.0.2     | org.axonframework.messaging.core.unitofwork       | unitofwork      | 11526                | 38914                      |       |
| 1  | axon-messaging-5.0.2     | org.axonframework.messaging.core                  | core            | 11118                | 35970                      |       |
| 2  | axon-common-5.0.2        | org.axonframework.common.configuration            | configuration   | 3366                 | 18028                      |       |
| 3  | axon-messaging-5.0.2     | org.axonframework.messaging.eventhandling         | eventhandling   | 3202                 | 10798                      |       |
| 4  | axon-common-5.0.2        | org.axonframework.common.annotation               | annotation      | 2196                 | 2438                       |       |
| 5  | axon-messaging-5.0.2     | org.axonframework.messaging.commandhandling       | commandhandling | 1463                 | 5131                       |       |
| 6  | axon-common-5.0.2        | org.axonframework.common.infra                    | infra           | 1377                 | 1666                       |       |
| 7  | axon-common-5.0.2        | org.axonframework.common                          | common          | 895                  | 2129                       |       |
| 8  | axon-messaging-5.0.2     | org.axonframework.messaging.queryhandling         | queryhandling   | 698                  | 2468                       |       |
| 9  | axon-messaging-5.0.2     | org.axonframework.messaging.core.annotation       | annotation      | 487                  | 1508                       |       |
| 10 | axon-messaging-5.0.2     | org.axonframework.messaging.eventhandling.proc... | token           | 425                  | 2147                       |       |
| 11 | axon-conversion-5.0.2    | org.axonframework.conversion                      | conversion      | 415                  | 1708                       |       |
| 12 | axon-messaging-5.0.2     | org.axonframework.messaging.eventstreaming        | eventstreaming  | 293                  | 833                        |       |
| 13 | axon-eventsourcing-5.0.2 | org.axonframework.eventsourcing.eventstore        | eventstore      | 221                  | 718                        |       |
| 14 | axon-messaging-5.0.2     | org.axonframework.messaging.tracing               | tracing         | 136                  | 478                        |       |
| 15 | axon-test-5.0.2          | org.axonframework.test.fixture                    | fixture         | 132                  | 359                        |       |
| 16 | axon-modelling-5.0.2     | org.axonframework.modelling                       | modelling       | 125                  | 282                        |       |
| 17 | axon-modelling-5.0.2     | org.axonframework.modelling.entity                | entity          | 115                  | 394                        |       |
| 18 | axon-modelling-5.0.2     | org.axonframework.modelling.entity.child          | child           | 89                   | 230                        |       |
| 19 | axon-messaging-5.0.2     | org.axonframework.messaging.eventhandling.proc... | segmenting      | 86                   | 662                        |       |

Table 1b

- Show the top 20 Java Packages including their sub-packages with the most incoming dependencies
- Set the property "incomingDependenciesIncludingSubpackages" on Package nodes.

|    | artifactName             | fullQualifiedPackageName                          | packageName     | incomingDependencies | incomingDependenciesWeight | incon |
|----|--------------------------|---------------------------------------------------|-----------------|----------------------|----------------------------|-------|
| 0  | axon-messaging-5.0.2     | org.axonframework.messaging.core                  | core            | 18039                | 59506                      |       |
| 1  | axon-messaging-5.0.2     | org.axonframework.messaging.core.unitofwork       | unitofwork      | 11278                | 38018                      |       |
| 2  | axon-common-5.0.2        | org.axonframework.common                          | common          | 7295                 | 23086                      |       |
| 3  | axon-common-5.0.2        | org.axonframework.common.configuration            | configuration   | 3205                 | 17368                      |       |
| 4  | axon-common-5.0.2        | org.axonframework.common.annotation               | annotation      | 2195                 | 2437                       |       |
| 5  | axon-messaging-5.0.2     | org.axonframework.messaging.eventhandling         | eventhandling   | 2081                 | 7484                       |       |
| 6  | axon-common-5.0.2        | org.axonframework.common.infra                    | infra           | 1363                 | 1592                       |       |
| 7  | axon-messaging-5.0.2     | org.axonframework.messaging.commandhandling       | commandhandling | 883                  | 3295                       |       |
| 8  | axon-messaging-5.0.2     | org.axonframework.messaging.core.annotation       | annotation      | 399                  | 1202                       |       |
| 9  | axon-conversion-5.0.2    | org.axonframework.conversion                      | conversion      | 363                  | 1527                       |       |
| 10 | axon-messaging-5.0.2     | org.axonframework.messaging.eventhandling.proc... | token           | 350                  | 1416                       |       |
| 11 | axon-messaging-5.0.2     | org.axonframework.messaging.queryhandling         | queryhandling   | 279                  | 897                        |       |
| 12 | axon-messaging-5.0.2     | org.axonframework.messaging.eventstreaming        | eventstreaming  | 241                  | 688                        |       |
| 13 | axon-messaging-5.0.2     | org.axonframework.messaging.eventhandling.proc... | processing      | 217                  | 685                        |       |
| 14 | axon-messaging-5.0.2     | org.axonframework.messaging.eventhandling.proc... | streaming       | 207                  | 655                        |       |
| 15 | axon-messaging-5.0.2     | org.axonframework.messaging.tracing               | tracing         | 107                  | 356                        |       |
| 16 | axon-eventsourcing-5.0.2 | org.axonframework.eventsourcing.eventstore        | eventstore      | 77                   | 208                        |       |
| 17 | axon-messaging-5.0.2     | org.axonframework.messaging.eventhandling.proc... | segmenting      | 74                   | 623                        |       |
| 18 | axon-modelling-5.0.2     | org.axonframework.modelling                       | modelling       | 74                   | 226                        |       |
| 19 | axon-messaging-5.0.2     | org.axonframework.messaging.core.conversion       | conversion      | 66                   | 225                        |       |

## Outgoing Dependencies

Outgoing dependencies are also denoted as "Fan-out", "Efferent Coupling" or "out-degree". These are the ones that are used by the listed package.

Code from other packages and libraries you're depending on (outgoing) might change over time. The more outgoing changes, the more likely and frequently code changes are needed. This involves time and effort which can be reduced by automation of tests and version updates. Automated tests are crucial to reveal updates, that change the behavior of the code unexpectedly ("fragile code"). As soon as more effort is required, keeping up becomes

difficult (“rigid code”). Not being able to use a newer version might not only restrict features, it can get problematic if there are security issues. This might force you to take “fast but ugly” solutions into account which further increases technical dept.

Since Java Packages are organized hierarchically, outgoing dependencies can be count for every package in isolation or by including all of its sub-packages. The latter one is done without top level packages like for example "org" or "org.company" by assuring that only packages are considered that have other packages or types in the same hierarchy level ("siblings").

## Table 2a

- Show the top 20 Java Packages with the most outgoing dependencies
- Set the "outgoingDependencies" properties on Package nodes.

|    | artifactName                         | fullQualifiedPackageName                          | packageName   | outgoingDependencies | outgoingDependenciesWeight | outgoin |
|----|--------------------------------------|---------------------------------------------------|---------------|----------------------|----------------------------|---------|
| 0  | axon-messaging-5.0.2                 | org.axonframework.messaging.core.configuration    | configuration | 2463                 | 15546                      |         |
| 1  | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | pooled        | 2321                 | 9770                       |         |
| 2  | axon-server-connector-5.0.2          | org.axonframework.axonserver.connector.event      | event         | 1282                 | 4029                       |         |
| 3  | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.eventstore.jpa    | jpa           | 1263                 | 4885                       |         |
| 4  | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.configuration     | configuration | 1237                 | 4801                       |         |
| 5  | axon-messaging-5.0.2                 | org.axonframework.messaging.core.interception     | interception  | 1084                 | 5770                       |         |
| 6  | axon-test-5.0.2                      | org.axonframework.test.fixture                    | fixture       | 972                  | 3461                       |         |
| 7  | axon-modelling-5.0.2                 | org.axonframework.modelling.entity.annotation     | annotation    | 858                  | 3265                       |         |
| 8  | axon-modelling-5.0.2                 | org.axonframework.modelling.entity                | entity        | 800                  | 2887                       |         |
| 9  | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | subscribing   | 749                  | 2715                       |         |
| 10 | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.eventstore        | eventstore    | 700                  | 2610                       |         |
| 11 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.autoconfig | autoconfig    | 658                  | 2833                       |         |
| 12 | axon-modelling-5.0.2                 | org.axonframework.modelling.configuration         | configuration | 585                  | 2279                       |         |
| 13 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling         | queryhandling | 533                  | 2104                       |         |
| 14 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling.dist... | distributed   | 513                  | 1412                       |         |
| 15 | axon-messaging-5.0.2                 | org.axonframework.messaging.monitoring.interce... | interception  | 500                  | 1550                       |         |
| 16 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling.inte... | interception  | 491                  | 1561                       |         |
| 17 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.anno... | annotation    | 478                  | 1359                       |         |
| 18 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling         | eventhandling | 471                  | 1197                       |         |
| 19 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.annotation       | annotation    | 464                  | 1548                       |         |

Table 2b

- Show the top 20 Java Packages including their sub-packages with the most outgoing dependencies
- Set the property "outgoingDependenciesIncludingSubpackages" on Package nodes.

|    | artifactName                         | fullQualifiedPackageName                          | packageName     | outgoingDependencies | outgoingDependenciesWeight | outgo |
|----|--------------------------------------|---------------------------------------------------|-----------------|----------------------|----------------------------|-------|
| 0  | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling         | eventhandling   | 671                  | 2124                       |       |
| 1  | axon-messaging-5.0.2                 | org.axonframework.messaging.core                  | core            | 402                  | 1431                       |       |
| 2  | axon-modelling-5.0.2                 | org.axonframework.modelling                       | modelling       | 381                  | 1124                       |       |
| 3  | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | processing      | 345                  | 1265                       |       |
| 4  | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing                   | eventsourcing   | 329                  | 1064                       |       |
| 5  | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling       | commandhandling | 315                  | 1014                       |       |
| 6  | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling         | queryhandling   | 314                  | 1122                       |       |
| 7  | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | streaming       | 274                  | 1006                       |       |
| 8  | axon-modelling-5.0.2                 | org.axonframework.modelling.entity                | entity          | 270                  | 856                        |       |
| 9  | axon-server-connector-5.0.2          | org.axonframework.axonserver.connector            | connector       | 253                  | 866                        |       |
| 10 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | pooled          | 231                  | 919                        |       |
| 11 | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.eventstore        | eventstore      | 194                  | 599                        |       |
| 12 | axon-modelling-5.0.2                 | org.axonframework.modelling.entity.annotation     | annotation      | 179                  | 681                        |       |
| 13 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.unitofwork       | unitofwork      | 176                  | 693                        |       |
| 14 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.interception     | interception    | 166                  | 587                        |       |
| 15 | axon-test-5.0.2                      | org.axonframework.test                            | test            | 160                  | 479                        |       |
| 16 | axon-server-connector-5.0.2          | org.axonframework.axonserver.connector.event      | event           | 142                  | 416                        |       |
| 17 | axon-test-5.0.2                      | org.axonframework.test.fixture                    | fixture         | 132                  | 472                        |       |
| 18 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.annotation       | annotation      | 124                  | 337                        |       |
| 19 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.autoconfig | autoconfig      | 121                  | 437                        |       |

## Instability

$$Instability = \frac{Outgoing\ Dependencies}{Outgoing\ Dependencies + Incoming\ Dependencies}$$

*Instability* is expressed as the ratio of the number of outgoing dependencies of a module (i.e., the number of packages that depend on it) to the total number of dependencies (i.e., the sum of incoming and outgoing dependencies).

Small values near zero indicate low *Instability*. With no outgoing but some incoming dependencies the *Instability* is zero which is denoted as maximally stable. Such code units are more rigid and difficult to change without impacting other parts of the system. If they are changed less because of that, they are considered stable.

Conversely, high values approaching one indicate high *Instability*. With some outgoing dependencies but no incoming ones the *Instability* is denoted as maximally unstable. Such code units are easier to change without affecting other modules, making them more flexible and less prone to cascading changes throughout the system. If they are changed more often because of that, they are considered unstable.

Since Java Packages are organized hierarchically, *Instability* can be calculated for every package in isolation or by including all of its sub-packages.

### Table 3a

- Show the top 20 Java Packages with the lowest *Instability*
- Set the property "instability" on Package nodes.

|    | artifactName          | fullQualifiedPackageName                          | packageName     | instability | instabilityTypes | instabilityInterfaces | instabilityF |
|----|-----------------------|---------------------------------------------------|-----------------|-------------|------------------|-----------------------|--------------|
| 0  | axon-common-5.0.2     | org.axonframework.common.annotation               | annotation      | 0.001364    | 0.021429         | 0.000000              |              |
| 1  | axon-messaging-5.0.2  | org.axonframework.messaging.core.unitofwork       | unitofwork      | 0.009283    | 0.080460         | 0.142857              |              |
| 2  | axon-common-5.0.2     | org.axonframework.common.infra                    | infra           | 0.019231    | 0.076336         | 0.090909              |              |
| 3  | axon-messaging-5.0.2  | org.axonframework.messaging.core                  | core            | 0.029758    | 0.163934         | 0.177083              |              |
| 4  | axon-common-5.0.2     | org.axonframework.common                          | common          | 0.033477    | 0.056478         | 0.000000              |              |
| 5  | axon-messaging-5.0.2  | org.axonframework.messaging.eventhandling.proc... | token           | 0.036281    | 0.121212         | 0.200000              |              |
| 6  | axon-conversion-5.0.2 | org.axonframework.conversion                      | conversion      | 0.048165    | 0.156250         | 0.250000              |              |
| 7  | axon-common-5.0.2     | org.axonframework.common.configuration            | configuration   | 0.074257    | 0.262626         | 0.295775              |              |
| 8  | axon-messaging-5.0.2  | org.axonframework.messaging.eventhandling         | eventhandling   | 0.128233    | 0.216495         | 0.392157              |              |
| 9  | axon-messaging-5.0.2  | org.axonframework.messaging.commandhandling       | commandhandling | 0.168750    | 0.263158         | 0.431818              |              |
| 10 | axon-common-5.0.2     | org.axonframework.common.util                     | util            | 0.200000    | 0.153846         | 0.000000              |              |
| 11 | axon-messaging-5.0.2  | org.axonframework.messaging.core.conversion       | conversion      | 0.238636    | 0.185185         | 0.571429              |              |
| 12 | axon-messaging-5.0.2  | org.axonframework.messaging.eventhandling.proc... | store           | 0.263158    | 0.190476         | 1.000000              |              |
| 13 | axon-messaging-5.0.2  | org.axonframework.messaging.eventstreaming        | eventstreaming  | 0.263819    | 0.320513         | 0.461538              |              |
| 14 | axon-messaging-5.0.2  | org.axonframework.messaging.monitoring            | monitoring      | 0.278481    | 0.195122         | 0.666667              |              |
| 15 | axon-common-5.0.2     | org.axonframework.common.lifecycle                | lifecycle       | 0.333333    | 0.444444         | 0.000000              |              |
| 16 | axon-messaging-5.0.2  | org.axonframework.messaging.tracing               | tracing         | 0.333333    | 0.320755         | 0.533333              |              |
| 17 | axon-update-5.0.2     | org.axonframework.update.api                      | api             | 0.379310    | 0.384615         | 0.000000              |              |
| 18 | axon-messaging-5.0.2  | org.axonframework.messaging.eventhandling.proc... | errorhandling   | 0.380952    | 0.384615         | 0.750000              |              |
| 19 | axon-test-5.0.2       | org.axonframework.test.server                     | server          | 0.400000    | 0.400000         | 0.000000              |              |

Table 3b

- Show the top 20 Java Packages including their sub-packages with the lowest *Instability*
- Set the property "instabilityIncludingSubpackages" on Package nodes.



|    | artifactName                         | fullQualifiedPackageName                          | packageName     | instability | instabilityTypes | instabilityInterfaces | instabilityF |
|----|--------------------------------------|---------------------------------------------------|-----------------|-------------|------------------|-----------------------|--------------|
| 0  | axon-common-5.0.2                    | org.axonframework.common.annotation               | annotation      | 0.000910    | 0.014493         | 0.000000              |              |
| 1  | axon-common-5.0.2                    | org.axonframework.common.infra                    | infra           | 0.003655    | 0.025210         | 0.000000              |              |
| 2  | axon-messaging-5.0.2                 | org.axonframework.messaging.core.unitofwork       | unitofwork      | 0.015366    | 0.050420         | 0.500000              |              |
| 3  | axon-common-5.0.2                    | org.axonframework.common.configuration            | configuration   | 0.021075    | 0.070175         | 0.425287              |              |
| 4  | axon-messaging-5.0.2                 | org.axonframework.messaging.core                  | core            | 0.021799    | 0.185590         | 0.391304              |              |
| 5  | axon-conversion-5.0.2                | org.axonframework.conversion                      | conversion      | 0.059585    | 0.146341         | 0.500000              |              |
| 6  | axon-common-5.0.2                    | org.axonframework.common.util                     | util            | 0.090909    | 0.100000         | 0.000000              |              |
| 7  | axon-messaging-5.0.2                 | org.axonframework.messaging.core.conversion       | conversion      | 0.131579    | 0.160000         | 0.500000              |              |
| 8  | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | token           | 0.152542    | 0.328358         | 0.500000              |              |
| 9  | axon-messaging-5.0.2                 | org.axonframework.messaging.eventstreaming        | eventstreaming  | 0.157343    | 0.204082         | 0.500000              |              |
| 10 | axon-common-5.0.2                    | org.axonframework.common.lifecycle                | lifecycle       | 0.166667    | 0.250000         | 0.000000              |              |
| 11 | axon-messaging-5.0.2                 | org.axonframework.messaging.tracing               | tracing         | 0.207407    | 0.300000         | 0.461538              |              |
| 12 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | errorhandling   | 0.230769    | 0.333333         | 0.500000              |              |
| 13 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.annotation       | annotation      | 0.237094    | 0.307692         | 0.421053              |              |
| 14 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling         | eventhandling   | 0.243823    | 0.515000         | 0.566265              |              |
| 15 | axon-common-5.0.2                    | org.axonframework.common.jdbc                     | jdbc            | 0.250000    | 0.266667         | 0.000000              |              |
| 16 | axon-common-5.0.2                    | org.axonframework.common.property                 | property        | 0.250000    | 0.400000         | 0.000000              |              |
| 17 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.util       | util            | 0.250000    | 0.333333         | 0.000000              |              |
| 18 | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling       | commandhandling | 0.262938    | 0.533333         | 0.527273              |              |
| 19 | axon-modelling-5.0.2                 | org.axonframework.modelling.repository            | repository      | 0.266667    | 0.333333         | 0.500000              |              |

## Abstractness

$$Abstractness = \frac{\text{abstract classes in category}}{\text{total number of classes in category}}$$

Package *Abstractness* is expressed as the ratio of the number of abstract classes and interfaces to the total number of classes of a package.

Zero *Abstractness* means that there are no abstract types or interfaces in the package. On the other hand, a value of one means that there are only abstract types.

Since Java Packages are organized hierarchically, *Abstractness* can be calculated for every package in isolation or by including all of its sub-packages.

Table 4a

- Show the top 30 packages with the lowest *Abstractness*
- Set the property "abstractness" on Package nodes.

|    | artifactName                         | fullQualifiedPackageName                          | packageName   | abstractness | numberAbstractTypes | numberTypes |
|----|--------------------------------------|---------------------------------------------------|---------------|--------------|---------------------|-------------|
| 0  | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.autoconfig | autoconfig    | 0.0          | 0                   | 35          |
| 1  | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot            | springboot    | 0.0          | 0                   | 22          |
| 2  | axon-server-connector-5.0.2          | org.axonframework.axonserver.connector.event      | event         | 0.0          | 0                   | 14          |
| 3  | axon-metrics-micrometer-5.0.2        | org.axonframework.extension.metrics.micrometer    | micrometer    | 0.0          | 0                   | 12          |
| 4  | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling.inte... | interception  | 0.0          | 0                   | 8           |
| 5  | axon-messaging-5.0.2                 | org.axonframework.messaging.core.timeout          | timeout       | 0.0          | 0                   | 8           |
| 6  | axon-update-5.0.2                    | org.axonframework.update.api                      | api           | 0.0          | 0                   | 6           |
| 7  | axon-messaging-5.0.2                 | org.axonframework.messaging.tracing.attributes    | attributes    | 0.0          | 0                   | 6           |
| 8  | axon-server-connector-5.0.2          | org.axonframework.axonserver.connector.command    | command       | 0.0          | 0                   | 6           |
| 9  | axon-messaging-5.0.2                 | org.axonframework.messaging.monitoring.interce... | interception  | 0.0          | 0                   | 5           |
| 10 | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling.in... | interception  | 0.0          | 0                   | 5           |
| 11 | axon-conversion-5.0.2                | org.axonframework.conversion.converter            | converter     | 0.0          | 0                   | 5           |
| 12 | axon-conversion-5.0.2                | org.axonframework.conversion.json                 | json          | 0.0          | 0                   | 5           |
| 13 | axon-common-5.0.2                    | org.axonframework.common.lifecycle                | lifecycle     | 0.0          | 0                   | 5           |
| 14 | axon-tracing-opentelemetry-5.0.2     | org.axonframework.extension.tracing.opentelemetry | opentelemetry | 0.0          | 0                   | 5           |
| 15 | axon-update-5.0.2                    | org.axonframework.update.detection                | detection     | 0.0          | 0                   | 4           |
| 16 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | jpa           | 0.0          | 0                   | 4           |
| 17 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.inte... | interception  | 0.0          | 0                   | 3           |
| 18 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | inmemory      | 0.0          | 0                   | 3           |
| 19 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.configuration... | reflection    | 0.0          | 0                   | 3           |
| 20 | axon-test-5.0.2                      | org.axonframework.test.server                     | server        | 0.0          | 0                   | 2           |
| 21 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.actuato... | axonserver    | 0.0          | 0                   | 2           |
| 22 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | annotation    | 0.0          | 0                   | 2           |
| 23 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.reflection       | reflection    | 0.0          | 0                   | 2           |
| 24 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.unitofwork.an... | annotation    | 0.0          | 0                   | 2           |
| 25 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.configuration    | configuration | 0.0          | 0                   | 2           |
| 26 | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.eventstore.jdbc   | jdbc          | 0.0          | 0                   | 2           |
| 27 | axon-update-5.0.2                    | org.axonframework.update.common                   | common        | 0.0          | 0                   | 1           |
| 28 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.util.jpa   | jpa           | 0.0          | 0                   | 1           |
| 29 | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling.retry | retry         | 0.0          | 0                   | 1           |

Table 4b

- Show the top 30 packages with the highest *Abstractness* and number of Java Types

|     | artifactName                         | fullQualifiedPackageName                          | packageName     | abstractness | numberAbstractTypes | numberTypes |
|-----|--------------------------------------|---------------------------------------------------|-----------------|--------------|---------------------|-------------|
| 142 | axon-common-5.0.2                    | org.axonframework.common.function                 | function        | 1.000000     | 2                   | 2           |
| 143 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.actuator   | actuator        | 1.000000     | 1                   | 1           |
| 144 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | streaming       | 1.000000     | 1                   | 1           |
| 145 | axon-common-5.0.2                    | org.axonframework.common.tx                       | tx              | 1.000000     | 1                   | 1           |
| 140 | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling.co... | configuration   | 0.750000     | 3                   | 4           |
| 141 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling.conf... | configuration   | 0.750000     | 3                   | 4           |
| 139 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.conf... | configuration   | 0.714286     | 10                  | 14          |
| 138 | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.annotation        | annotation      | 0.700000     | 7                   | 10          |
| 137 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.unitofwork.tr... | transaction     | 0.600000     | 3                   | 5           |
| 135 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.util       | util            | 0.571429     | 4                   | 7           |
| 136 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling.anno... | annotation      | 0.571429     | 4                   | 7           |
| 134 | axon-modelling-5.0.2                 | org.axonframework.modelling.repository            | repository      | 0.555556     | 5                   | 9           |
| 133 | axon-modelling-5.0.2                 | org.axonframework.modelling.configuration         | configuration   | 0.538462     | 7                   | 13          |
| 128 | axon-common-5.0.2                    | org.axonframework.common.jdbc                     | jdbc            | 0.500000     | 6                   | 12          |
| 129 | axon-common-5.0.2                    | org.axonframework.common.annotation               | annotation      | 0.500000     | 2                   | 4           |
| 130 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.conv... | conversion      | 0.500000     | 1                   | 2           |
| 131 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling.gateway | gateway         | 0.500000     | 1                   | 2           |
| 132 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.conversion       | conversion      | 0.500000     | 1                   | 2           |
| 127 | axon-common-5.0.2                    | org.axonframework.common.configuration            | configuration   | 0.478261     | 22                  | 46          |
| 125 | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling       | commandhandling | 0.473684     | 9                   | 19          |
| 126 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling         | eventhandling   | 0.473684     | 9                   | 19          |
| 124 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.interception.... | annotation      | 0.454545     | 5                   | 11          |
| 123 | axon-test-5.0.2                      | org.axonframework.test.fixture                    | fixture         | 0.451613     | 14                  | 31          |
| 122 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventstreaming        | eventstreaming  | 0.437500     | 7                   | 16          |
| 120 | axon-modelling-5.0.2                 | org.axonframework.modelling.entity.child          | child           | 0.428571     | 6                   | 14          |
| 121 | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing                   | eventsourcing   | 0.428571     | 3                   | 7           |
| 119 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling         | queryhandling   | 0.416667     | 10                  | 24          |
| 116 | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling.di... | distributed     | 0.400000     | 4                   | 10          |
| 117 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.repl... | annotation      | 0.400000     | 4                   | 10          |
| 118 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.gateway | gateway         | 0.400000     | 2                   | 5           |

## Table 4c

- Show the top 30 packages including their sub-packages with the highest package depth and lowest *Abstractness*
- Set the property "abstractnessIncludingSubpackages" on Package nodes.

|    | artifactName                         | fullQualifiedPackageName                          | packageName   | abstractness | numberAbstractTypes | numberTypes | maxSub |
|----|--------------------------------------|---------------------------------------------------|---------------|--------------|---------------------|-------------|--------|
| 0  | axon-metrics-micrometer-5.0.2        | org.axonframework.extension.metrics.micrometer    | micrometer    | 0.0          | 0                   | 13          |        |
| 1  | axon-messaging-5.0.2                 | org.axonframework.messaging.core.configuration    | configuration | 0.0          | 0                   | 5           |        |
| 2  | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.autoconfig | autoconfig    | 0.0          | 0                   | 35          |        |
| 3  | axon-server-connector-5.0.2          | org.axonframework.axonserver.connector.event      | event         | 0.0          | 0                   | 14          |        |
| 4  | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling.inte... | interception  | 0.0          | 0                   | 8           |        |
| 5  | axon-messaging-5.0.2                 | org.axonframework.messaging.core.timeout          | timeout       | 0.0          | 0                   | 8           |        |
| 6  | axon-update-5.0.2                    | org.axonframework.update.api                      | api           | 0.0          | 0                   | 6           |        |
| 7  | axon-messaging-5.0.2                 | org.axonframework.messaging.tracing.attributes    | attributes    | 0.0          | 0                   | 6           |        |
| 8  | axon-server-connector-5.0.2          | org.axonframework.axonserver.connector.command    | command       | 0.0          | 0                   | 6           |        |
| 9  | axon-messaging-5.0.2                 | org.axonframework.messaging.monitoring.interce... | interception  | 0.0          | 0                   | 5           |        |
| 10 | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling.in... | interception  | 0.0          | 0                   | 5           |        |
| 11 | axon-conversion-5.0.2                | org.axonframework.conversion.converter            | converter     | 0.0          | 0                   | 5           |        |
| 12 | axon-conversion-5.0.2                | org.axonframework.conversion.json                 | json          | 0.0          | 0                   | 5           |        |
| 13 | axon-common-5.0.2                    | org.axonframework.common.lifecycle                | lifecycle     | 0.0          | 0                   | 5           |        |
| 14 | axon-tracing-opentelemetry-5.0.2     | org.axonframework.extension.tracing.opentelemetry | opentelemetry | 0.0          | 0                   | 5           |        |
| 15 | axon-update-5.0.2                    | org.axonframework.update.detection                | detection     | 0.0          | 0                   | 4           |        |
| 16 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | jpa           | 0.0          | 0                   | 4           |        |
| 17 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.inte... | interception  | 0.0          | 0                   | 3           |        |
| 18 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | inmemory      | 0.0          | 0                   | 3           |        |
| 19 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.configuration... | reflection    | 0.0          | 0                   | 3           |        |
| 20 | axon-test-5.0.2                      | org.axonframework.test.server                     | server        | 0.0          | 0                   | 2           |        |
| 21 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.actuato... | axonserver    | 0.0          | 0                   | 2           |        |
| 22 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | annotation    | 0.0          | 0                   | 2           |        |
| 23 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.reflection       | reflection    | 0.0          | 0                   | 2           |        |
| 24 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.unitofwork.an... | annotation    | 0.0          | 0                   | 2           |        |
| 25 | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.eventstore.jdbc   | jdbc          | 0.0          | 0                   | 2           |        |
| 26 | axon-update-5.0.2                    | org.axonframework.update.common                   | common        | 0.0          | 0                   | 1           |        |
| 27 | axon-spring-boot-                    | org.axonframework.extension.springboot.util.jpa   | jpa           | 0.0          | 0                   | 1           |        |

|    | artifactName                  | fullQualifiedPackageName                          | packageName | abstractness | numberAbstractTypes | numberTypes | maxSub |
|----|-------------------------------|---------------------------------------------------|-------------|--------------|---------------------|-------------|--------|
|    | autoconfigure-5.0.2           |                                                   |             |              |                     |             |        |
| 28 | axon-messaging-5.0.2          | org.axonframework.messaging.commandhandling.retry | retry       | 0.0          | 0                   | 1           |        |
| 29 | axon-metrics-micrometer-5.0.2 | org.axonframework.extension.metrics.micrometer... | reservoir   | 0.0          | 0                   | 1           |        |

Table 4d

- Show the top 30 packages including their sub-packages with the highest package depth and highest *Abstractness*

|     | artifactName                         | fullQualifiedPackageName                          | packageName     | abstractness | numberAbstractTypes | numberTypes | max5 |
|-----|--------------------------------------|---------------------------------------------------|-----------------|--------------|---------------------|-------------|------|
| 115 | axon-common-5.0.2                    | org.axonframework.common.function                 | function        | 1.000000     | 2                   | 2           |      |
| 116 | axon-common-5.0.2                    | org.axonframework.common.tx                       | tx              | 1.000000     | 1                   | 1           |      |
| 113 | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling.co... | configuration   | 0.750000     | 3                   | 4           |      |
| 114 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling.conf... | configuration   | 0.750000     | 3                   | 4           |      |
| 112 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.conf... | configuration   | 0.714286     | 10                  | 14          |      |
| 111 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.unitofwork.tr... | transaction     | 0.600000     | 3                   | 5           |      |
| 110 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling.anno... | annotation      | 0.571429     | 4                   | 7           |      |
| 109 | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.annotation        | annotation      | 0.562500     | 9                   | 16          |      |
| 108 | axon-modelling-5.0.2                 | org.axonframework.modelling.repository            | repository      | 0.555556     | 5                   | 9           |      |
| 107 | axon-modelling-5.0.2                 | org.axonframework.modelling.configuration         | configuration   | 0.538462     | 7                   | 13          |      |
| 101 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.util       | util            | 0.500000     | 4                   | 8           |      |
| 102 | axon-common-5.0.2                    | org.axonframework.common.jdbc                     | jdbc            | 0.500000     | 6                   | 12          |      |
| 103 | axon-common-5.0.2                    | org.axonframework.common.annotation               | annotation      | 0.500000     | 2                   | 4           |      |
| 104 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.conv... | conversion      | 0.500000     | 1                   | 2           |      |
| 105 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling.gateway | gateway         | 0.500000     | 1                   | 2           |      |
| 106 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.conversion       | conversion      | 0.500000     | 1                   | 2           |      |
| 100 | axon-common-5.0.2                    | org.axonframework.common.configuration            | configuration   | 0.478261     | 22                  | 46          |      |
| 99  | axon-messaging-5.0.2                 | org.axonframework.messaging.core.interception.... | annotation      | 0.454545     | 5                   | 11          |      |
| 98  | axon-test-5.0.2                      | org.axonframework.test.fixture                    | fixture         | 0.451613     | 14                  | 31          |      |
| 97  | axon-messaging-5.0.2                 | org.axonframework.messaging.eventstreaming        | eventstreaming  | 0.437500     | 7                   | 16          |      |
| 96  | axon-modelling-5.0.2                 | org.axonframework.modelling.entity.child          | child           | 0.428571     | 6                   | 14          |      |
| 93  | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling.di... | distributed     | 0.400000     | 4                   | 10          |      |
| 94  | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.repl... | annotation      | 0.400000     | 4                   | 10          |      |
| 95  | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.gateway | gateway         | 0.400000     | 2                   | 5           |      |
| 92  | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling         | queryhandling   | 0.380952     | 24                  | 63          |      |
| 91  | axon-modelling-5.0.2                 | org.axonframework.modelling                       | modelling       | 0.376344     | 35                  | 93          |      |
| 89  | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling       | commandhandling | 0.375000     | 24                  | 64          |      |

|    | artifactName         | fullQualifiedPackageName                          | packageName | abstractness | numberAbstractTypes | numberTypes | maxS |
|----|----------------------|---------------------------------------------------|-------------|--------------|---------------------|-------------|------|
| 90 | axon-messaging-5.0.2 | org.axonframework.messaging.commandhandling.ga... | gateway     | 0.375000     | 3                   | 8           |      |
| 88 | axon-messaging-5.0.2 | org.axonframework.messaging.eventhandling.anno... | annotation  | 0.368421     | 7                   | 19          |      |
| 86 | axon-messaging-5.0.2 | org.axonframework.messaging.core.unitofwork       | unitofwork  | 0.363636     | 8                   | 22          |      |

## Distance from the main sequence

The *main sequence* is a imaginary line that represents a good compromise between *Abstractness* and *Instability*. A high distance to this line may indicate problems. For example is very *stable* (rigid) code with low abstractness hard to change.

Read more details on that in [OO Design Quality Metrics](#) and [Calculate metrics](#).

### Table 5a

- Show the top 30 packages with the highest distance from the "main sequence"



|    | artifactName                         | fullQualifiedName                                 | name          | distance | abstractness | instability | elementsCount |
|----|--------------------------------------|---------------------------------------------------|---------------|----------|--------------|-------------|---------------|
| 0  | axon-messaging-5.0.2                 | org.axonframework.messaging.tracing.attributes    | attributes    | NaN      | 0.0          | NaN         | 6             |
| 1  | axon-conversion-5.0.2                | org.axonframework.conversion.converter            | converter     | NaN      | 0.0          | NaN         | 5             |
| 2  | axon-messaging-5.0.2                 | org.axonframework.messaging.core.reflection       | reflection    | NaN      | 0.0          | NaN         | 2             |
| 3  | axon-common-5.0.2                    | org.axonframework.common.function                 | function      | NaN      | 1.0          | NaN         | 2             |
| 4  | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.actuator   | actuator      | NaN      | 1.0          | NaN         | 1             |
| 5  | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling.retry | retry         | NaN      | 0.0          | NaN         | 1             |
| 6  | axon-metrics-micrometer-5.0.2        | org.axonframework.extension.metrics.micrometer... | reservoir     | NaN      | 0.0          | NaN         | 1             |
| 7  | axon-common-5.0.2                    | org.axonframework.common.io                       | io            | NaN      | 0.0          | NaN         | 1             |
| 8  | axon-common-5.0.2                    | org.axonframework.common.digest                   | digest        | NaN      | 0.0          | NaN         | 1             |
| 9  | axon-test-5.0.2                      | org                                               | org           | NaN      | 0.0          | NaN         | 0             |
| 10 | axon-test-5.0.2                      | org.axonframework                                 | axonframework | NaN      | 0.0          | NaN         | 0             |
| 11 | axon-update-5.0.2                    | org                                               | org           | NaN      | 0.0          | NaN         | 0             |
| 12 | axon-update-5.0.2                    | org.axonframework                                 | axonframework | NaN      | 0.0          | NaN         | 0             |
| 13 | axon-spring-boot-autoconfigure-5.0.2 | org                                               | org           | NaN      | 0.0          | NaN         | 0             |
| 14 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework                                 | axonframework | NaN      | 0.0          | NaN         | 0             |
| 15 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension                       | extension     | NaN      | 0.0          | NaN         | 0             |
| 16 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.service    | service       | NaN      | 0.0          | NaN         | 0             |
| 17 | axon-messaging-5.0.2                 | org                                               | org           | NaN      | 0.0          | NaN         | 0             |
| 18 | axon-messaging-5.0.2                 | org.axonframework                                 | axonframework | NaN      | 0.0          | NaN         | 0             |
| 19 | axon-messaging-5.0.2                 | org.axonframework.messaging                       | messaging     | NaN      | 0.0          | NaN         | 0             |
| 20 | axon-conversion-5.0.2                | org                                               | org           | NaN      | 0.0          | NaN         | 0             |
| 21 | axon-conversion-5.0.2                | org.axonframework                                 | axonframework | NaN      | 0.0          | NaN         | 0             |
| 22 | axon-metrics-micrometer-5.0.2        | org                                               | org           | NaN      | 0.0          | NaN         | 0             |
| 23 | axon-metrics-micrometer-5.0.2        | org.axonframework                                 | axonframework | NaN      | 0.0          | NaN         | 0             |
| 24 | axon-metrics-micrometer-5.0.2        | org.axonframework.extension                       | extension     | NaN      | 0.0          | NaN         | 0             |
| 25 | axon-metrics-micrometer-5.0.2        | org.axonframework.extension.metrics               | metrics       | NaN      | 0.0          | NaN         | 0             |
| 26 | axon-modelling-5.0.2                 | org                                               | org           | NaN      | 0.0          | NaN         | 0             |
| 27 | axon-modelling-5.0.2                 | org.axonframework                                 | axonframework | NaN      | 0.0          | NaN         | 0             |
| 28 | axon-eventsourcing-5.0.2             | org                                               | org           | NaN      | 0.0          | NaN         | 0             |
| 29 | axon-eventsourcing-5.0.2             | org.axonframework                                 | axonframework | NaN      | 0.0          | NaN         | 0             |

Table 5b

- Show the top 30 packages including their sub-packages with the highest distance from the "main sequence"

|    | artifactName                         | fullQualifiedName                                 | name           | distance | abstractness | instability | elementsCount |
|----|--------------------------------------|---------------------------------------------------|----------------|----------|--------------|-------------|---------------|
| 0  | axon-common-5.0.2                    | org.axonframework.common.lifecycle                | lifecycle      | 0.833333 | 0.000000     | 0.166667    | 5             |
| 1  | axon-conversion-5.0.2                | org.axonframework.conversion                      | conversion     | 0.773748 | 0.166667     | 0.059585    | 30            |
| 2  | axon-common-5.0.2                    | org.axonframework.common.infra                    | infra          | 0.746345 | 0.250000     | 0.003655    | 8             |
| 3  | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | token          | 0.718425 | 0.129032     | 0.152542    | 31            |
| 4  | axon-messaging-5.0.2                 | org.axonframework.messaging.core                  | core           | 0.693797 | 0.284404     | 0.021799    | 218           |
| 5  | axon-update-5.0.2                    | org.axonframework.update.api                      | api            | 0.684211 | 0.000000     | 0.315789    | 6             |
| 6  | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling.co... | configuration  | 0.675000 | 0.750000     | 0.925000    | 4             |
| 7  | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling.conf... | configuration  | 0.675000 | 0.750000     | 0.925000    | 4             |
| 8  | axon-test-5.0.2                      | org.axonframework.test.server                     | server         | 0.666667 | 0.000000     | 0.333333    | 2             |
| 9  | axon-messaging-5.0.2                 | org.axonframework.messaging.core.unitofwork       | unitofwork     | 0.620998 | 0.363636     | 0.015366    | 22            |
| 10 | axon-common-5.0.2                    | org.axonframework.common.util                     | util           | 0.575758 | 0.333333     | 0.090909    | 6             |
| 11 | axon-messaging-5.0.2                 | org.axonframework.messaging.tracing               | tracing        | 0.529435 | 0.263158     | 0.207407    | 19            |
| 12 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling.anno... | annotation     | 0.523810 | 0.571429     | 0.952381    | 7             |
| 13 | axon-common-5.0.2                    | org.axonframework.common.configuration            | configuration  | 0.500664 | 0.478261     | 0.021075    | 46            |
| 14 | axon-update-5.0.2                    | org.axonframework.update.detection                | detection      | 0.500000 | 0.000000     | 0.500000    | 4             |
| 15 | axon-update-5.0.2                    | org.axonframework.update.common                   | common         | 0.500000 | 0.000000     | 0.500000    | 1             |
| 16 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.util.jpa   | jpa            | 0.500000 | 0.000000     | 0.500000    | 1             |
| 17 | axon-common-5.0.2                    | org.axonframework.common.annotation               | annotation     | 0.499090 | 0.500000     | 0.000910    | 4             |
| 18 | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.annotation        | annotation     | 0.474781 | 0.562500     | 0.912281    | 16            |
| 19 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.annotation       | annotation     | 0.442906 | 0.320000     | 0.237094    | 50            |
| 20 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling         | eventhandling  | 0.440388 | 0.315789     | 0.243823    | 171           |
| 21 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | errorhandling  | 0.435897 | 0.333333     | 0.230769    | 3             |
| 22 | axon-common-5.0.2                    | org.axonframework.common.property                 | property       | 0.416667 | 0.333333     | 0.250000    | 9             |
| 23 | axon-modelling-5.0.2                 | org.axonframework.modelling.configuration         | configuration  | 0.415174 | 0.538462     | 0.876712    | 13            |
| 24 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventstreaming        | eventstreaming | 0.405157 | 0.437500     | 0.157343    | 16            |
| 25 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.actuato... | axonserver     | 0.400000 | 0.000000     | 0.600000    | 2             |
| 26 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling.gateway | gateway        | 0.400000 | 0.500000     | 0.900000    | 2             |
| 27 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.conf... | configuration  | 0.371589 | 0.714286     | 0.657303    | 14            |
| 28 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.conversion       | conversion     | 0.368421 | 0.500000     | 0.131579    | 2             |
| 29 | axon-common-5.0.2                    | org.axonframework.common.jpa                      | jpa            | 0.365385 | 0.250000     | 0.384615    | 4             |

## Abstractness vs. Instability Plot with "Main Sequence" line as reference

- Plot *Abstractness* vs. *Instability* of all packages
- Draw the "main sequence" as dashed green diagonal line
- Scale the packages by the number of types they contain
- Color the packages by their distance to the "main sequence" (blue=near, red=far)

Figure 5a - Packages without their sub-packages

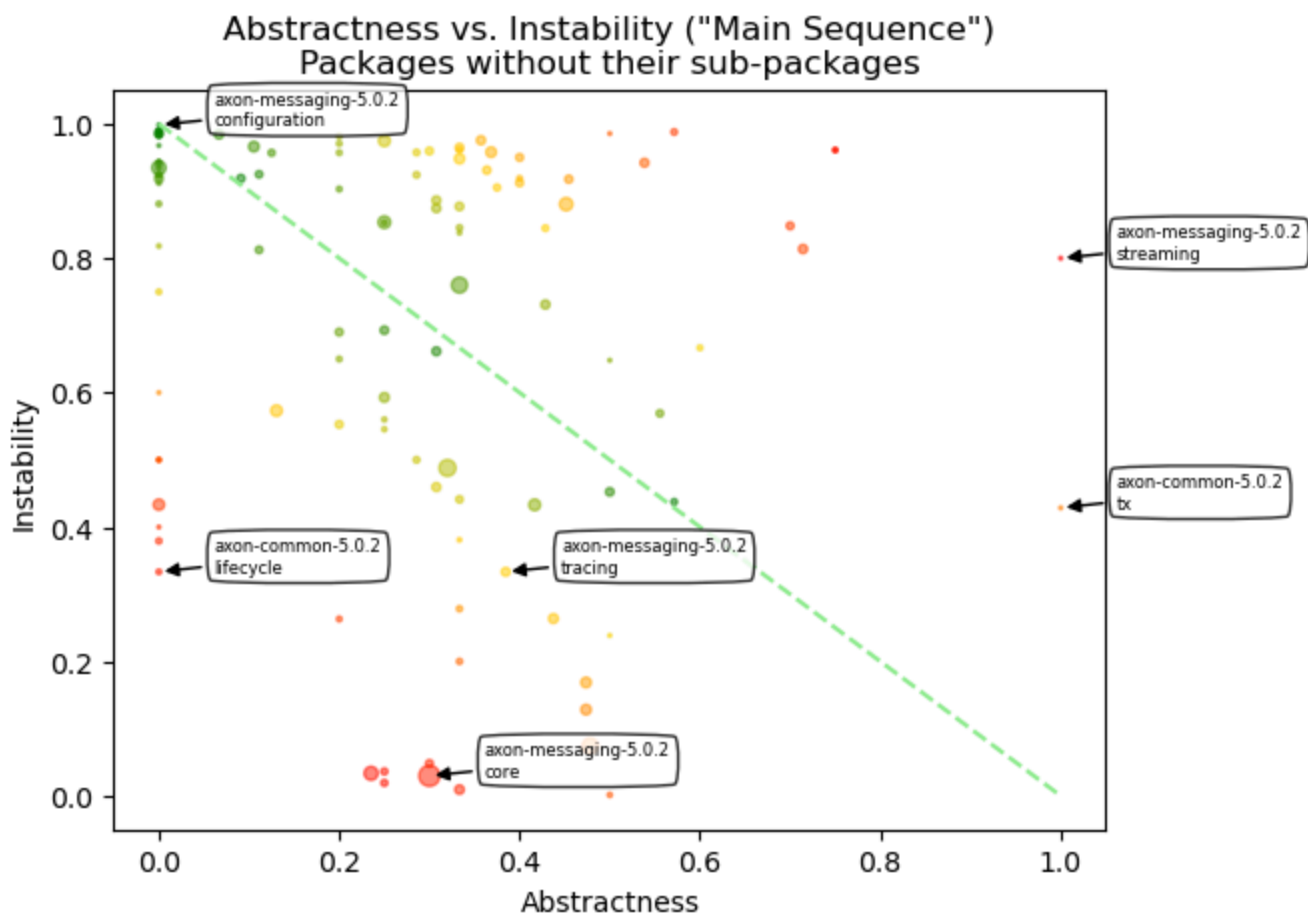


Figure 5b - Packages including their sub-packages

Abstractness vs. Instability ("Main Sequence")  
Packages including their sub-packages

